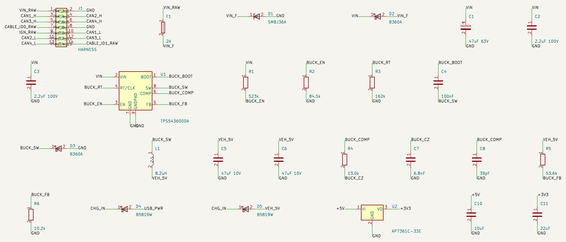
VEHICLE INPUT: protected 9–36V → 5V buck; battery-backed rails below



J1: Harness: 1 VIN, 2/8 GND, 3–6 CAN1–4 H, 10–13 CAN1–4 L, 9 IGN, 7/14 cable ID

D1: TVS: load dump / jump start; reverse polarity blows F1 via D1

D2: Reverse polarity block

U1: Values from TI TPS54360 datasheet 5V design example (Fig. 34)

R2: UVLO: start 8V, stop 6.25V

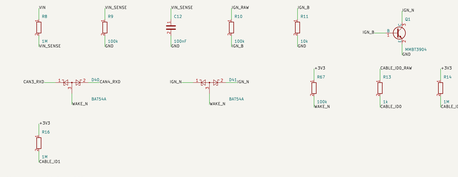
R3: fsw ~600 kHz

D3: Catch diode

D4: USB bench supply only through J99: use a rated 5V source, not an unregulated host port

D5: Vehicle/USB input OR: battery cannot backfeed either input

VIN / IGNITION / CABLE-ID SENSE



R9: VIN/11 → ADC: 36V = 3.27V

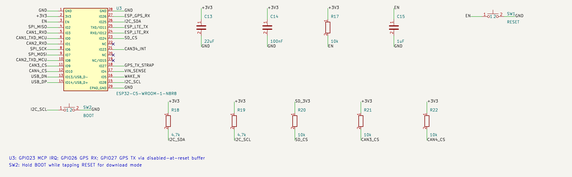
Q1: IGN > ~6.5V pulls IGN_N low (6–60V safe)

D6/D7: WAKE_N goes low on CAN3/CAN4 bus activity (transceiver in standby) or ignition on

R67: Deep-sleep wake sources: GPIO3 CAN1_RXD, GPIO1 CAN2_RXD, GPIO5 WAKE_N

R16: Cable grounds ID0/ID1 to identify itself (1M ≈ 3 uA in sleep)

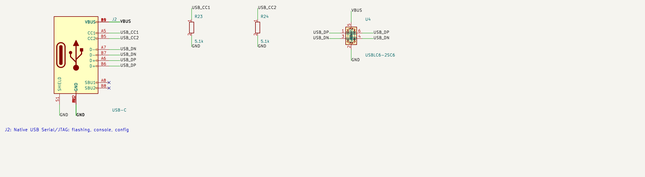
MCU: ESP32-C5-WROOM-1 (2x native CAN-FD, Wi-Fi 6 + Wi-Fi Aware)



U3: GPIO23 MCP IRQ; GPIO26 GPS RX; GPIO27 GPS TX via disabled-at-reset buffer

SW2: Hold BOOT while tapping RESET for download mode

USB-C (native USB Serial/JTAG)



CAN1 (native CAN-FD 0) — OBD gateway pins 6/14

U5: STB has internal pull-up; transceiver boots in standby

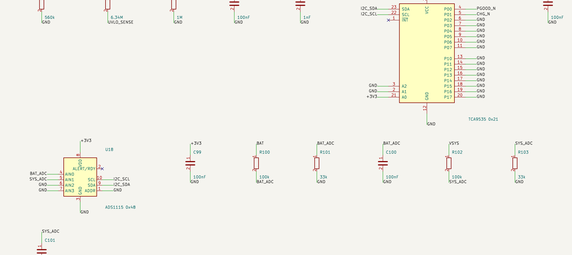
JP1: Fit ONLY when the logger is a bus end

JP3: TX ENABLE: no jumper = channel physically cannot transmit

CAN2 (native CAN-FD 1)

U6: STB has internal pull-up; transceiver boots in standby

CAN3 (MCP2518FD #1) + shared 40MHz clock



U7: STB has internal pull-up; transceiver boots in standby

U9: INT isolated by AND gate; byte-write iOCON: STB through XSTBY

Y1: One clock for both MCP2518FD; expander can stop it in sleep

U19: Active-low interrupts combined without tying outputs together

CAN4 (MCP2518FD #2)

U8: STB has internal pull-up; transceiver boots in standby

U10: INT isolated by AND gate; byte-write iOCON: STB through XSTBY

microSD (SPI)

J3: SPI mode, shares the bus with the MCP2518FDs

Q3: SD card power switch: card is unpowered in sleep (idle cards can draw ~1 mA)

R65: Keep the SPI bus defined while the MCU sleeps

IO EXPANDER, STATUS LEDS, MARK BUTTON

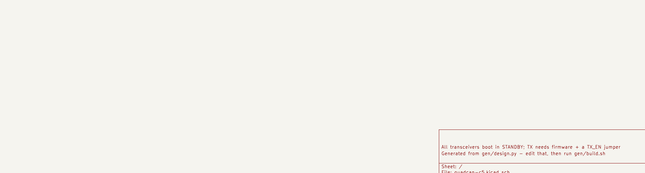
U11: Powers up as inputs: STB lines float high = transceivers in standby

SW3: Drops an event marker into the log

RTC + CR1220 backup



GPS (optional) NEO-M9N 25Hz + active antenna bias-T



U13: Optional: leave U13 + bias-T unpopulated to rely on the phone's GPS

Q2: GPS power switch

R93: Bias-T per NEO-M9N integration manual; limits short-circuit current

R94: Blinks once per second with a fix

U20: OE high while GPS off; boot GPIO27 sees only buffer input

EXPANSION: LTE header, Qwiic

J5: LTE Cat-1 add-on (e.g. SIMCom A7670 carrier with its own regulator, fed from VIN)

MECHANICAL

BATTERY

JP9: Fit only for a >=1A 5V bench supply; USB data works without jumper

J7: Verify pack polarity: 1 plus, 2 NTC, 3 minus: protected 1S Li-ion only

U14: ITEMP/TMR intentionally open: default termination/timers; TS must reach actual pack NTC

R90: Nominal 100mA charge, worst ICHST ~110mA

R91: Nominal ~0.996 input limit: total board plus charging

U15: 5V boost with true output disconnect: enables only above hardware UVLO

L3: Verify saturation >= peak switch current over temperature

R95: 0.595V reference; 4.950V nominal PWM output

U16: Hardware hysteresis: -3.15V off / -3.66V restart, nominal; not pack protection

U17: All pins input; unused inputs externally grounded

U18: 4/ND battery / 4/ND system, both divided by 4.03; ALERT unused

FLAGS

All transceivers boot in STANDBY; TX needs firmware + a TX_EN jumper

Generated from gen/design.py — edit that, then run gen/build.sh

Sheet: /

File: quadcan-t5.kicad_sch

Title: QuadCAN—C5: 4—channel CAN—FD logger

Size: A2 | Date: 2026-11-13 | **Rev: 1.1—battery—prototype**

KiCad 8.0.4 9.0.2 | Id: 1/1